The Critical-Line Principle for Engineered Rationality: Why "Half-Truth" Is the Optimal Rigidity of Logic (A Riemann-Hypothesis Lens on Name–Dao–Logic)

Abstract

In the Name–Dao–Logic framework, "truth" is not metaphysical; it is the subset of semantic structure that a logic can enforce **profitably** under environmental volatility, coordination pressure, and enforcement cost. We formalize this as an **anchoring fraction** ρ (empirically proxied by AB-fixness) measuring how much of naming and policy is forced into cross-observer and cross-time agreement, and propose the **Critical-Line Principle (CLP)**: long-run viability typically peaks in a mid-rigidity regime, often near a "half-structured / half-residual" boundary ρ ≈ 1/2. To motivate why "1/2" is a natural design locus rather than numerology, we use a Riemann Hypothesis lens: RH places all nontrivial zeros on the symmetry boundary Re(s)=1/2, and (conditional) prime-counting errors admit a canonical √x-scale envelope (e.g., O(√x log x)), exemplifying a sharp separation between controllable trend and irreducible residual fluctuation. (Wikipedia) We further leverage proven "partial criticality" results—Conrey's ≥2/5 and Bui–Conrey–Young's >41% of zeros on the critical line—as a parable for engineered rationality: meaningful anchoring can be **provably secured** even when global idealization remains unproven, legitimizing "partial truth" as an explicit design target. (American Institute of Mathematics) Finally, we give a concrete non-absoluteness mechanism via anisotropic 3-axis representational frames, showing how apparent logical universality can arise from hidden coordinate choices at the naming layer.

---

1. Introduction: logic as engineered rigidity, not timeless law

Modern discussions often treat logic as a timeless evaluator of propositions. In the Name–Dao–Logic framework, logic is instead an **engineered rigidity layer**: it is a protocol that constrains naming (N) and policy (D) so that agents can coordinate, compress experience, and survive under environmental change. "Truth," in this view, is not a metaphysical primitive; it is the **subset of structure worth enforcing** given drift, cost, and multi-observer disagreement.

1.1 What the reader already knows (assumed from *Name, Dao, and Logic*)

We assume the reader accepts the central evaluation criterion: a logic is good when it is **viable** in an environment, not when it satisfies an abstract ideal of purity.

(1.1) V(L; E) = E[S(N_L, D_L; E)].

Here E is the environment (including drift and volatility), S(·) is an accumulated survival/return functional, and (N_L, D_L) are the Name and Dao induced (or constrained) by logic L.

The framework also introduces AB-fixness as a **tunable rigidity control**: how strongly a system forces agreement across observers and across time.

(1.2) $0 \le A_B(L) \le 1$.

Intuitively:

- $A_B(L) \approx 1$ means strong cross-observer + cross-time invariance (high rigidity).
- $A_B(L) \approx 0$ means weak invariance (high flexibility, but poor coordination and weak accumulation).

This paper treats $A_B(L)$ not as a moral virtue but as a **design parameter**.

1.2 What this paper adds

This paper adds three things to the Name–Dao–Logic program.

(i) A design rule for "useful truth."
We formalize the idea that the *useful part of truth* tends to stabilize near a **mid-rigidity regime**, rather than at maximal rigidity ($A_B \approx 1$) or maximal flexibility ($A_B \approx 0$). We call this the **Half-Truth design rule**: the system enforces a stable invariant core while deliberately retaining an adaptive residual.

(ii) A bridge to the Riemann critical line (conceptual, not a proof of RH).
The Riemann Hypothesis (RH) asserts that all nontrivial zeros of the zeta function lie on the critical line $\mathrm{Re}(s)=1/2$. (Clay Mathematics Institute)
This paper uses the critical line as a **canonical mathematical archetype** for a boundary between "controllable structure" and "irreducible residual fluctuation," motivating why "1/2" naturally appears as an optimal rigidity boundary in engineered rational systems. No claim is made that RH implies our principle, or that our principle proves RH.

(1.3) RH (statement used here): $\zeta(s)=0$ (nontrivial) $\Rightarrow \mathrm{Re}(s)=1/2$. (Clay Mathematics Institute)

(iii) A concrete demonstration that logic is not absolute via anisotropic 3-axis frames.
To make "non-absoluteness" tangible (not rhetorical), we introduce a minimal model where representational geometry is anisotropic: changing the underlying 3-axis weighting changes what counts as "similar," hence changes category boundaries under naming. This shows how "logical universality" can be an artifact of a hidden shared coordinate choice.

1.3 Reader-facing promise (what is falsifiable / testable)

The paper makes a testable prediction: **in adaptive, drifting environments, best long-run performance occurs in a mid-rigidity band**, not at the extremes. The key quantity is the interaction between enforced agreement and environmental volatility.

Let σ_E denote a measurable volatility/drift scale of the environment (any operational proxy is acceptable, e.g., regime-switch frequency, distribution shift magnitude, or prediction error under a fixed baseline).

(1.4) $A_B(L) \cdot \sigma_E \in [\alpha_{min}, \alpha_{max}]$.

Interpretation: viability degrades when the product $A_B \cdot \sigma_E$ is too small (insufficient anchoring relative to drift) or too large (over-rigidity under drift). The interval $[\alpha_{min}, \alpha_{max}]$ is not assumed universal; it is **learnable and environment-dependent**, and can be estimated empirically.

We further commit to a simple "shape claim" that can be tested even in toy simulations: for an appropriate rigidity proxy ρ (often $\rho \approx A_B$), performance exhibits a single interior maximum.

(1.5) $\exists\ \rho^* \in (0,1)$ such that $J(\rho)$ is maximized at $\rho = \rho^*$.
(1.6) In the symmetric complementarity regime (defined later), $\rho^* = 1/2$.

Concretely, the experiments will evaluate:

- single-agent adaptation under drift (performance vs rigidity),
- multi-agent coordination and disagreement costs (A_B vs task success),
- frame-perturbation tests where the representational geometry shifts (anisotropic-axis intervention).

If the observed optimum systematically collapses to an extreme (near 0 or near 1), or if performance is monotone in rigidity, the core design rule of this paper is falsified.

---

2. Background tools: Name–Dao–Logic primitives (minimal recap)

This section fixes notation and isolates only the primitives needed for the "Half-Truth / Critical-Line" argument. We follow *Name, Dao, and Logic* in treating logic as an engineered constraint-layer that shapes how agents **name** the world and **act** in it, under environmental drift and coordination costs.

2.1 Core maps (only what we need)

We model a "world instance" (raw situation) as $w \in W$, where W is the space of pre-named inputs (observations, contexts, histories, sensor streams, etc.). Naming compresses w into a semantic state $x \in X$. Action selection maps x into an action $a \in A$.

(2.1) $N : W \to X$.
(2.2) $D : X \to A$.

Two clarifications matter for later sections:

- **Logic L constrains N and D indirectly.** L does not need to "compute" N or D; rather, it supplies constraints (consistency rules, update discipline, admissible inferences, and utility pressures) that shape how N_L and D_L are formed or selected.
- **Degrees of freedom live inside N and D.** "Rigidity" will mean "how much of N and D is forced to be stable across observers and across time," as measured by AB-fixness.

2.2 Survival / return functional

We evaluate an (N, D) pair inside an environment E by an accumulated return functional $S(N, D; E)$, which may represent survival probability, reward, fitness proxy, or any operational performance metric.

(2.3) $S(N, D; E) = \sum_{t=0}^{T} \gamma^t R(x_t, a_t; E)$.

Typical trajectory semantics (informal but sufficient here):

- $x_t = N(w_t)$ is the named state at time t.
- $a_t = D(x_t)$ is the action chosen at time t.

- $R(\cdot)$ is the environment-dependent payoff (or survival increment).
- $\gamma \in (0,1]$ is a discount factor (or horizon control).

Later, the key point is not the exact form of R, but that **S couples stability and adaptivity**: stable naming/policies help accumulation, while flexible naming/policies help adaptation under drift.

2.3 AB-fixness as a measurable statistic (one concrete instantiation)

To quantify "how much truth is enforced," we need a rigidity observable. AB-fixness $A_B(L)$ is a concrete, measurable statistic capturing cross-observer agreement (and, by extension, cross-time stability when time-indexed).

Let A and B be two observers/agents (or the same agent at two times, two checkpoints, two regimes). Let π_E be a distribution over world instances induced by the environment.

(2.4) $A_B(L) = E_{w \sim \pi_E}[I(N_A(w)=N_B(w) \wedge D_A(N_A(w))=D_B(N_B(w)))]$.

Interpretation:

- The indicator $I(\cdot)$ returns 1 when both **naming** and **policy** agree on w (after naming), and 0 otherwise.
- $A_B(L)$ is therefore the fraction of encountered situations where "the same world collapses into the same named state and triggers the same action," under the regime defined by L.

This definition is intentionally strict (exact equality). In experiments, one may use softened versions (e.g., similarity thresholds on X, action-equivalence classes in A), but (2.4) is the clean baseline that makes the paper's claims falsifiable.

2.4 The "cost of truth": enforcement + complexity

A central engineering constraint is that **more rigidity is never free**. Enforcing stable invariants across observers/time incurs costs: complexity of the naming scheme, maintenance of consistency, arbitration overhead, delayed adaptation, and brittleness under drift.

We package this into a semantic-Lagrangian-style objective. The sign convention below treats L_{sem} as something to minimize.

(2.5) $L_{sem}(N, D, L; E) = -V(L; E) + \lambda_N C(N) + \lambda_L Cost(L)$.

Where:

- $V(L; E)$ is viability (often $V = E[S(N_L, D_L; E)]$, consistent with Section 1).
- $C(N)$ is the representational/ontological cost of naming (e.g., description length, model size, number of categories, update complexity).
- $Cost(L)$ is the enforcement/coordination cost of the logic layer (e.g., rule-checking overhead, arbitration cost, constraint violations, rigidity penalties).
- $\lambda_N, \lambda_L \geq 0$ are tradeoff weights.

This is the mechanical reason "absolute truth" is not a default optimum: pushing A_B toward 1 typically increases $C(N)$ and $Cost(L)$ and can reduce V under drift. Conversely, pushing A_B toward 0 collapses coordination and destroys accumulation. The rest of the paper will show how this tradeoff naturally creates an **interior optimum** for rigidity, and why "1/2" is the canonical

balance point in the symmetric regime.

---

3. The Critical-Line Principle: defining "Half-Truth" as a design variable

This section introduces a single control variable—the **anchoring fraction**—to turn "truth" from a metaphysical claim into an **engineering knob**. The key move is to treat logic L as a mechanism that decides *how much* of the semantic representation must be **invariant** across observers and across time, versus how much must remain **free** to absorb drift, context, and novelty.

3.1 Definition: anchoring fraction (the "truth we actually enforce")

Let $\rho \in [0,1]$ denote the fraction of semantic degrees of freedom that are forced into cross-observer and cross-time agreement.

(3.1) $\rho := A_B(L)$.

In the simplest scalar setting, we directly identify ρ with AB-fixness as defined in (2.4). This makes ρ operational: it is measurable as an agreement rate over world instances sampled from π_E.

In many systems, rigidity is not one-dimensional. Different subsystems can have different "agreement pressures" (e.g., naming vs policy, synchronization vs temporal stability). We therefore also allow a vector rigidity control:

(3.2) $\vec{\rho} := (A_{sync}, A_{time}, A_{name}, A_{policy}, \ldots)$.

To connect the vector to the scalar statements below, define an aggregation functional $g(\cdot)$ that maps a rigidity profile into an effective anchoring level:

(3.3) $\rho_{eff} := g(\vec{\rho})$, with $\rho_{eff} \in [0,1]$.

A simple default is a weighted mean (weights represent which rigidity dimensions dominate viability in a given environment):

(3.4) $\rho_{eff} = \sum_{i=1}^{k} w_i \rho_i$, *with* $w_i \geq 0$ *and* $\sum_{i=1}^{k} w_i = 1$.

From here onward, "ρ" may be read as either the scalar AB-fixness (3.1) or an effective anchoring level ρ_{eff} (3.3).

3.2 Definition: "useful truth" vs "residual truth"

We define **truth** in this paper as "what the system enforces as invariant structure," not as a timeless property of propositions.

- **Useful truth** is the enforced invariant core: the part of representation and policy that (i) survives coarse-graining, (ii) supports coordination and accumulation, and (iii) remains stable under a finite enforcement budget.
- **Residual truth** is the deliberately unforced remainder: context, renaming freedom, exploratory hypotheses, strategy diversity, and locally-tuned exceptions—everything that must remain flexible so the system can adapt without shattering.

To make the split concrete, suppose the agent's representation decomposes into an invariant

core u and a residual context v:

(3.5) $\Phi(w) = (u, v)$, where $u \in U_{core}$ and $v \in U_{res}$.

Then a "high-anchoring" regime forces agreement primarily through u, while v remains free to vary across situations (or across observers) without being treated as contradiction:

(3.6) $N(w) = N(u, v) = (N_{core}(u), N_{res}(v))$.

The anchoring fraction ρ can be read as "how much of the system's effective decision-relevant structure is routed through the invariant core rather than the residual."

A minimal (and measurable) proxy of this split is to define a set of invariants I_ρ that the logic attempts to preserve, and treat "useful truth" as those invariants:

(3.7) $T_{use}(\rho) := I_\rho$ (enforced invariants).
(3.8) $T_{res}(\rho) :=$ total_structure $\setminus I_\rho$ (allowed-to-drift structure).

The enforcement budget (2.5) implies that $|I_\rho|$ cannot grow without cost; therefore, increasing ρ is never free.

3.3 Main claim: Critical-Line Principle (CLP)

We now state the core principle.

Critical-Line Principle (CLP). In adaptive systems under drift and coordination costs, optimal long-run viability often occurs near a mid-rigidity regime:

(3.9) $\rho^* \approx 1/2$.

This is stated as a **design principle**, not a theorem about all logics. Its engineering justification is the familiar "two-failure-modes" shape:

- ρ too high ⇒ brittle invariants, high enforcement cost, slow adaptation ⇒ V drops.
- ρ too low ⇒ under-coordination, no accumulation, drifting Names/Daos ⇒ V drops.

To give CLP a compact mathematical spine (still consistent with "principle, not proof"), we introduce a toy viability proxy that captures the complementarity between anchoring and slack.

Model 3.A (Complementarity proxy). Let the net benefit from enforcement be proportional to having both (i) invariant structure and (ii) adaptive slack, while costs rise with rigidity:

(3.10) $J(\rho) := \kappa \cdot \rho \cdot (1-\rho) - c(\rho)$, with $\kappa > 0$.

A simplest first pass is a linear cost $c(\rho)=c_1\rho+c_0$ with $c_1 \geq 0$, giving:

(3.11) $J(\rho) = \kappa\rho(1-\rho) - c_1\rho - c_0$.

This produces an interior optimum near 1/2 whenever the complementarity term dominates the marginal cost over the relevant range. In the symmetric, low-cost limit ($c_1 \approx 0$), the maximizer is exactly 1/2:

(3.12) $dJ/d\rho = \kappa(1-2\rho)$.
(3.13) $dJ/d\rho = 0 \Rightarrow \rho^* = 1/2$.

More generally, asymmetry in how much the environment "demands" invariance versus slack shifts the optimum away from 1/2. A convenient parametric generalization is:

(3.14) $J(\rho) := \kappa \cdot \rho^p \cdot (1-\rho)^q - c_0$, with $p,q > 0$.
(3.15) $\rho^* = p / (p+q)$.

This is how the paper uses "1/2" correctly: **1/2 is the symmetric critical point** (p=q), while real environments may produce ρ^* slightly above or below 1/2 depending on whether coordination pressure (p) or adaptivity pressure (q) dominates.

CLP, therefore, is the claim that many real systems operate near the symmetric regime often enough that "half-truth" becomes a stable design attractor—especially when agents must both coordinate and remain robust to drift.

3.4 What "1/2" means in this paper (preview)

The "half" in Half-Truth does **not** mean "half the sentences are true" or any probabilistic claim about propositions.

It means a **structural split**:

- roughly half of the system's effective semantic degrees of freedom are forced into invariant form (coordination-ready, accumulable, cross-observer stable),
- and roughly half are preserved as adaptive slack (contextual, revisable, drift-tolerant, innovation-capable).

In later sections, we will motivate why "1/2" is a canonical boundary by using the Riemann critical line as a mathematical archetype for "optimal boundary between controllable structure and irreducible residual," and we will show a concrete non-absoluteness mechanism via anisotropic 3-axis frames (where changing the representational geometry changes what invariants are even definable).

---

4. The Riemann Hypothesis lens: why 1/2 is a natural boundary

This section uses RH as a **mathematical archetype** of a "critical boundary": a place where pushing a global error uniformly below a √·-type barrier would require structural changes that RH explicitly disallows. The role of RH here is conceptual scaffolding for CLP—not a proof.

4.1 The minimal RH fact we use

RH asserts that all **nontrivial zeros** of the Riemann zeta function lie on the **critical line**.

(4.0) $\zeta(s)=0$ (nontrivial) $\Rightarrow \mathrm{Re}(s)=1/2$. (Clay Mathematics Institute)

A related fact (used for "why 1/2 is special") is that the functional equation maps $s \leftrightarrow 1-s$, so the symmetry line is $\mathrm{Re}(s)=1/2$ (often written $\sigma=1/2$).

4.2 Why the critical line behaves like an "optimal error boundary"

The prime-counting function $\pi(x)$ fluctuates around smooth approximations such as $\mathrm{li}(x)$. RH implies a sharp, global control of this fluctuation, often presented as a $\sqrt{x}$-scale bound (up to logs).

(4.1) $\pi(x) - \mathrm{li}(x) = O(\sqrt{x} \log x)$. (Wikipedia)

A more "boundary-shaped" way to state the same phenomenon is: the **best possible exponent** in bounds of the form x^{β} (up to logs) is governed by how far right zeros can sit in the critical strip. One standard formulation is:

(4.2) If $\beta := \sup\{ \mathrm{Re}(\rho) : \zeta(\rho)=0, 0<\mathrm{Re}(\rho)<1 \}$, then $\pi(x) - \mathrm{li}(x) = O(x^{\beta} \log x)$. (Wikipedia)

Under RH, $\beta=1/2$, recovering (4.1). (Wikipedia)

Interpretational mapping (explicitly an analogy).
We read $\sqrt{x}$ scaling as the canonical signature of:

- a **structured trend** (li(x) captures the leading-order "law"), plus
- an **irreducible residual** whose typical envelope behaves like a square-root barrier (up to log factors).

That "trend + residual with a $\sqrt{\cdot}$ barrier" is the mathematical template we borrow for *Half-Truth*: enforce a stable core, but accept that a residual must remain "uncompressed" (adaptive / fluctuating) if you want global robustness.

4.3 The "half" as a symmetry / balance point (no mysticism, just structure)

There are two structural reasons "1/2" is not arbitrary in the RH story:

(i) Symmetry of the analytic system.
The functional equation pairs s with $1-s$, making the midpoint $\mathrm{Re}(s)=1/2$ the symmetry axis of the critical strip.
So "1/2" is the natural *balance line* between the two sides of the strip.

(ii) Boundary between improvable and non-improvable uniform error exponents.
Using (4.2), demanding a uniform improvement such as

(4.3) $\pi(x) - \mathrm{li}(x) = O(x^{1/2-\varepsilon})$ (ignoring logs)

would force $\beta \le 1/2-\varepsilon$, i.e., it would require that **all** nontrivial zeros lie strictly to the left of the critical line. RH asserts they are on the line ($\beta=1/2$), so it forbids that kind of uniform exponent improvement. (Wikipedia)

Template (trend + oscillation).
Write any "counting" observable generically as:

(4.4) $\mathrm{Count}(x) = \mathrm{Trend}(x) + \mathrm{Osc}(x)$.

RH tells us: Trend(x) can be globally smooth, but Osc(x) has a natural $\sqrt{\cdot}$ envelope determined by the horizontal location of zeros—hence a canonical "critical boundary" at exponent 1/2.

This is exactly the structure CLP needs: a principled reason why "trying to enforce everything" (drive residual to 0 uniformly) is not only costly, but structurally blocked in the archetypal case.

4.4 What the RH lens does *not* claim

To avoid category errors, we make the non-claims explicit:

- No claim that RH proves CLP (the rigidity optimum $\rho^* \approx 1/2$ is an engineering principle about viable systems, not a theorem of analytic number theory).
- No claim that CLP implies RH.

- The actual claim is narrower: RH provides a **canonical example** where "1/2 is the boundary between what can be made uniformly small and what cannot," because the best achievable global error exponent is pinned to $\beta=1/2$ when zeros lie on the critical line. (Wikipedia)

5. The proven ≥ 0.4 anchor: "partial criticality" as a non-absolute truth demonstration

This section supplies a *rigorous* numerical foothold for the paper's central theme: even when the ideal "critical boundary" (the full RH picture) is not globally established, one can still prove **nontrivial partial alignment** with the critical line. We repurpose that fact as a parable for engineered rationality: "truth" can be **partially secured** (as a guaranteed anchoring fraction) without being absolute.

5.1 The Conrey 2/5 theorem as a rigidity floor

Let $N(T)$ be the number of nontrivial zeros $\rho = \beta+i\gamma$ of $\zeta(s)$ with $0 < \gamma \le T$ (counted with multiplicity), and let $N_0(T)$ be the number of such zeros *on the critical line* $\beta=1/2$.

(5.1) $\kappa := \liminf_{T\to\infty} N_0(T) / N(T)$.

Conrey (1989) proved that at least two-fifths of the zeros are (in fact, simple and) on the critical line, which implies the lower bound:

(5.2) $\kappa \ge 2/5$. (American Institute of Mathematics)

For this paper, the key point is not the mollifier technology, but the **logic-of-proof pattern**: the "ideal boundary" is unproven globally, yet a meaningful fraction of the global structure is already forced to align with it.

5.2 Later improvements (optional, brief)

Bui–Conrey–Young strengthened the guaranteed proportion on the line to **more than 41%** (often summarized as $\kappa > 0.41$, with explicit constants given in the paper). (American Institute of Mathematics)

We will treat this as an existence proof that the "anchoring fraction" can be pushed upward by better constructions, even without resolving RH.

5.3 How to use "0.4" inside Name–Dao–Logic (core conceptual move)

Mathematical parable (partial criticality).
Conrey's $\kappa \ge 2/5$ result (and later $\kappa > 0.41$) shows a situation where:

- the **target structure** ("all zeros on the line") is not yet proven, but
- we can still **prove a baseline fraction** of the system is already constrained to the target boundary.

Translation to engineered rationality.
In Name–Dao–Logic terms, interpret κ as a proxy for a *provable anchoring floor*: even when you cannot justify (or afford) an ideal rigidity choice like $\rho\approx 1/2$ globally, you may still be able to guarantee a minimum viable anchoring fraction.

(5.3) $\rho \geq \rho_{floor}$, where ρ_{floor} is provable under given assumptions.

The rhetorical payoff is sharp: **truth is not absolute, yet it is not arbitrary**. It can be *partially locked in* by the environment, by resource bounds, or by coordination constraints—exactly the way κ is partially locked in by analytic constraints and method.

5.4 Formalizing "partial anchoring" in the framework

We now express "partial criticality" as an explicit design constraint.

Let B denote an enforcement budget (computational cost, arbitration cost, rule-checking overhead, or any operational "truth enforcement" budget consistent with Section 2.4). Let σ_E denote an environment volatility/drift scale (Section 1.3). Then define feasible rigidity bounds:

(5.4) $\rho_{min}(E, B) \leq \rho \leq \rho_{max}(E, B)$.

Interpretation:

- $\rho_{min}(E,B)$: the minimum anchoring needed to prevent Name/Dao drift from destroying accumulation and coordination.
- $\rho_{max}(E,B)$: the maximum anchoring you can maintain before enforcement cost and brittleness under drift collapse viability.

To connect with the falsifiable "band" claim introduced earlier, we also enforce the viability-band constraint:

(5.5) $\alpha_{min} \leq \rho \cdot \sigma_E \leq \alpha_{max}$.

Now "half-truth" becomes a **target** inside a constrained region, rather than a dogma. A clean way to state the operational rule is:

(5.6) $\rho_{target} := 1/2$.
(5.7) $\rho_{design} := \text{clamp}(\rho_{target}, \rho_{min}(E,B), \rho_{max}(E,B))$.

where $\text{clamp}(y, a, b) := \min(\max(y,a), b)$.

This makes the logical role of "≥ 0.4" precise in the paper's language:

- Conrey/BCY tell us that in a canonical hard system (ζ zeros), **a nontrivial fraction is provably anchored** to the critical boundary. (American Institute of Mathematics)
- In engineered systems, this corresponds to having **provable or enforceable rigidity floors** (ρ_{min}) even when the "ideal" ($\rho \approx 1/2$) is not achievable everywhere.

In short: **partial criticality is the mathematically respectable version of "truth is non-absolute but still engineerable."**

6. What exactly are the "degrees of freedom" inside 1/2?

Section 3 introduced ρ as "how much semantic structure we force into cross-observer, cross-time agreement." This section makes that statement concrete: **what exactly is being forced, what is being left free, and how to measure the split**.

6.1 Decomposing degrees of freedom in Name–Dao–Logic terms

We separate the degrees of freedom (DoF) that a system can choose into three buckets, each with a different failure mode when over-constrained.

(i) Ontological DoF (Name).
These are choices inside $N : W \to X$, including:

- granularity (how many states/categories),
- boundary placement (what counts as "same" vs "different"),
- compression rate (how much detail is discarded).

A compact way to view this is: naming selects an equivalence relation $\sim$ on W, then X is the quotient space $W/\sim$.

(6.0) $w_1 \sim w_2 \Leftrightarrow N(w_1) = N(w_2)$.

Over-rigidity here produces brittle categories; under-rigidity produces fragmentation and loss of accumulation.

(ii) Policy DoF (Dao).
These are choices inside $D : X \to A$, including:

- action parameters,
- mode-switching rules (which "Dao" is active),
- exploration vs exploitation schedules.

Under-rigidity yields inconsistent behavior across agents/time; over-rigidity yields slow adaptation and poor robustness.

(iii) Enforcement DoF (Logic).
These are choices inside L that determine:

- which constraints are checked,
- how violations are penalized,
- where rigidity is local vs global,
- update schedules (when exceptions are allowed, when re-naming is permitted).

This is the "truth enforcement budget" control surface: logic can enforce strongly in some subsystems and weakly in others.

6.2 A concrete "half-truth" model: invariant core + adaptive sheath

To make "half" non-mystical, we model semantics as a **direct-sum split**: a core that must be stable, plus a residual that is explicitly allowed to drift.

(6.1) $X = X_{core} \oplus X_{residual}$.

Interpretation:

- X_{core} carries invariants you expect to persist across observers/time (coordination-ready structure).

- X_residual carries context, exceptions, novelty, drift-handling, exploratory hypotheses (adaptive slack).

A helpful operational picture is to define two projections:

(6.2) P_core : X → X_core.
(6.3) P_res : X → X_residual.
(6.4) x = P_core(x) ⊕ P_res(x).

Then "enforced truth" means: logic mostly forces agreement on P_core(x), while tolerating disagreement (or time drift) in P_res(x). In words:

- "You must agree on the core."
- "You may disagree on the sheath."

That alone is already enough to explain why "truth is non-absolute": disagreements can be real, but they are *designed to live in the residual*, not treated as global contradiction.

A concrete instantiation (naming).
Let the naming map factor through the split:

(6.5) N(w) = (N_core(w), N_res(w)), with N_core(w) ∈ X_core and N_res(w) ∈ X_residual.

A "Half-Truth" regime enforces high agreement on N_core and allows flexibility in N_res.

6.3 Connecting to AB-fixness vector form

Because "agreement" has multiple dimensions (naming, policy, temporal stability), we use a rigidity vector:

(6.6) $\vec{\rho}$ = (A_name, A_policy, A_time).

A clean, measurable set of components is:

(6.7) A_name = E_{w ~ π_E}[I(N_A(w) = N_B(w))].
(6.8) A_policy = E_{w ~ π_E}[I(D_A(N_A(w)) = D_B(N_B(w)))].
(6.9) A_time = E_{w ~ π_E}[I(N_t(w) = N_{t+Δ}(w))] (or a softened similarity version).

Then define an effective anchoring level (used when the paper states "ρ ≈ 1/2"):

(6.10) ρ_eff = w_1A_name + w_2A_policy + w_3A_time, with w_i ≥ 0 and w_1+w_2+w_3=1.

This explains the most important subtlety:

"ρ ≈ 1/2" does not mean each component equals 1/2.
It may mean allocations such as:

- A_name high (stable ontology),
- A_policy medium (some shared tactics),
- A_time low (explicit drift tolerance),
  while the weighted average stays near 1/2.

In practice this is often what you want: keep names stable enough to coordinate and accumulate, while allowing time-local reinterpretation and policy adaptation.

6.4 Operational proxies (so the paper is not purely philosophical)

To estimate the split empirically, you want **agreement metrics** (cross-agent) plus **drift metrics** (across time), separately for Name and Dao.

(i) Cross-agent label agreement (Name).

- Simple match rate (directly estimates A_name in (6.7)).
- If you need chance-corrected agreement, use Cohen-style κ on labels (optional).

(6.11) Â_name = (1/n) ∑_{i=1}^{n} I(N_A(w_i) = N_B(w_i)).

(ii) Cross-agent action agreement (Dao).

- Same idea on actions, possibly after mapping actions into equivalence classes if multiple actions are "functionally identical."

(6.12) Â_policy = (1/n) ∑_{i=1}^{n} I(D_A(N_A(w_i)) = D_B(N_B(w_i))).

(iii) Temporal drift (Name and Dao stability).
Track how the naming distribution changes over time:

(6.13) Drift_name(t, t+Δ) = D_div(P_t(N(w)), P_{t+Δ}(N(w))).

Where D_div can be KL, Jensen–Shannon, total variation, or any divergence you can estimate reliably.

Similarly for actions:

(6.14) Drift_policy(t, t+Δ) = D_div(P_t(D(N(w))), P_{t+Δ}(D(N(w)))).

(iv) Core vs residual agreement (directly testing the "half-truth split").
If you implement the decomposition (6.5), you can measure agreement separately on core and residual channels:

(6.15) Â_core = (1/n) ∑ I(N_core,A(w_i) = N_core,B(w_i)).
(6.16) Â_res = (1/n) ∑ I(N_res,A(w_i) = N_res,B(w_i)).

A true "Half-Truth" design typically aims for:

(6.17) Â_core high, Â_res moderate/low, with ρ_eff in a mid-band.

This section pins down what "degrees of freedom inside 1/2" actually are: **they are the parts of naming, policy, and enforcement that you deliberately choose not to freeze**, so the system can survive drift without turning every context change into a logical crisis.

7. The anisotropic 3-axis demonstration: how "absolute logic" can be a coordinate illusion

This section gives a concrete mechanism for "logic is not absolute" that does **not** rely on philosophical relativism. The mechanism is purely representational: **change the geometric frame used for compression**, and what counts as "the same thing," "a stable invariant," or even "a contradiction" can change—because those judgments are made *after* naming.

7.1 Motivation: "logic seems absolute" when the frame is hidden

Logic feels timeless when everyone implicitly shares the same representational geometry—i.e., the same notion of distance, similarity, and feature salience. In that case:

- different observers tend to cluster world instances in the same way,
- the same invariants (stable categories) emerge,
- and the same inference rules appear "obviously correct."

The "absoluteness" is often the **hidden consensus of a shared frame**, not a property of logic in isolation.

7.2 A minimal anisotropic 3D model (mathematically explicit, but lightweight)

Let the internal representation of a situation be $x \in R^3$. Define a weighted norm using a positive diagonal metric:

(7.1) $||x||^2_G = x^T G x$, $G = \mathrm{diag}(g_1, g_2, g_3)$, with $g_i > 0$.

Define the induced distance:

(7.2) $\mathrm{Dist}_G(a,b) = ||a-b||_G = \sqrt{((a-b)^T G (a-b))}$.

Isotropic case.
If $g_1=g_2=g_3$, then Dist_G is rotationally symmetric: similarity judgments do not depend on orientation. Category boundaries tend to be stable under reparameterizations.

Anisotropic case.
If g_1,g_2,g_3 are unequal, then some axes are "more expensive" than others. Two representations that look close under one metric can look far under another. This changes:

- nearest-neighbor structure,
- cluster assignments,
- and therefore the *ontology* induced by naming.

7.3 Mapping to Name–Dao–Logic

Naming is compression: it maps raw instances into discrete or low-dimensional semantic states by a geometry-dependent assignment rule. A minimal template is nearest-prototype naming.

Let $\Phi(w) \in R^3$ be an embedding of world instance w. Let $\{\mu_k\}$ be category prototypes (centroids) representing named states x_k.

(7.3) $N_G(w) = \mathrm{argmin}_{k} \mathrm{Dist}_G(\Phi(w), \mu_k)$.

Key consequence: if two observers A and B use different metrics G_A and G_B (or different axis weightings), then even with the same Φ and the same prototypes μ_k, they can assign different names:

(7.4) $N_{G_A}(w) \neq N_{G_B}(w)$ for some w.

Now observe what "contradiction" means in practice: logic operates over *named propositions*. A proposition p typically means "$x \in P$" for some subset $P \subseteq X$ of named states. But since naming depends on G, the same natural-language predicate can denote different subsets under different frames:

(7.5) p_G := “N_G(w) ∈ P”.
(7.6) p_{G′} := “N_{G′}(w) ∈ P”.

So an apparent cross-observer contradiction can arise even when each observer is internally consistent:

- Observer A: p_G is true (because w maps into P under G).
- Observer B: p_{G′} is false (because w maps outside P under G′).

Nothing mystical happened: the two observers are not disagreeing inside the same proposition space; they are using **different induced proposition spaces** because their Name maps differ.

This is exactly why “absolute logic” can be a coordinate illusion: what looks like a logical clash may be a **frame clash** at the naming layer.

Where “Half-Truth” matters.
A robust system should expect occasional frame shifts (drift, new sensors, new tasks). Under CLP, you stabilize a core (X_core) and allow an adaptive sheath (X_residual). A frame change can then be absorbed primarily by updating the residual—without collapsing the whole logic.

7.4 The “geocentric vs heliocentric” style pivot (as a logic pivot)

The classic pivot is not just “a new theory,” but a change in representational commitments: the same observational stream becomes compressible under a different coordinate choice.

Structured case-study template.

- Same observations: a sequence O(t) of planetary positions over time.
- Two coordinate commitments:
  - Frame F_geo: Earth-centered naming and invariants.
  - Frame F_helio: Sun-centered naming and invariants.

Under F_geo, certain patterns (e.g., retrograde motion) become “anomalies” that require additional machinery to preserve assumed invariants. Under F_helio, the same pattern becomes a simple relational consequence of relative motion, so what was treated as “exception logic” becomes ordinary.

In Name–Dao–Logic terms:

- The *data* w did not change.
- The naming map did: N_geo(w) and N_helio(w) carve the observation space differently.
- Therefore, which invariants are “natural” (and which rules seem “obvious”) changed with the frame.

This is precisely what the framework recommends when a system hits persistent logical strain: **update Name (ontology) to resolve apparent logical failures**, instead of forcing ever more rigid rules on a mis-specified naming layer.

(7.7) If repeated violations concentrate in X_residual, update N (and possibly Φ, G) before hardening L.

In other words: the “geocentric → heliocentric” move is a vivid instance of the general design lesson: when logic looks broken, it is often the *coordinate system of naming* that needs to move —so that the logic can become simple again.

8. From principle to AGI architecture: a “Half-Truth Controller”

This section turns CLP from a conceptual principle into an implementable architectural component: a controller that **tunes rigidity** (how much truth is enforced as invariant) rather than treating rigidity as a fixed assumption.

8.1 Design goal

The goal is to implement a module that **dynamically selects the anchoring fraction** ρ (or the rigidity vector $\vec{\rho}$) in response to environmental volatility, coordination demands, and enforcement budget. Instead of “logic = fixed rules,” we treat logic as a **controlled resource** that can be allocated.

Design target: keep the system in a mid-rigidity regime whenever possible, and only harden or loosen constraints when evidence suggests drift or coordination pressure has changed.

(8.0) $\rho_t = \text{Controller}(\hat{\sigma}_{E,t}, \text{budget}_t, \text{disagreement}_t, \text{task}_u)$.

Here $\hat{\sigma}_{E,t}$ is an estimated drift scale, disagreement_t is a measured cross-agent disagreement proxy, and task_u denotes the current context/subsystem.

8.2 Control surface: schedules over contexts

A practical AGI does not require uniform rigidity across all subsystems. Instead, rigidity is best treated as a **schedule** indexed by context u (domain, subsystem, task mode, risk tier, or timescale).

Let k_L be an enforcement strength (penalty weight, rule-checking strictness, arbitration intensity, or constraint temperature). Then:

(8.1) $k_L = k_L(u)$, where u is context / subsystem.

This implements “Half-Truth” as a *distribution* of rigidity across the system:

- high $k_L(u)$ in modules where shared invariants are essential (e.g., safety constraints, accounting identities, stable object permanence),
- low $k_L(u)$ where exploration and reinterpretation are valuable (e.g., hypothesis generation, creative planning, novel domains).

To connect $k_L(u)$ to anchoring fraction, define a monotone response mapping:

(8.2) $\rho(u) = f(k_L(u))$, with $f'(\cdot) \geq 0$ and $\rho(u) \in [0,1]$.

If you want a concrete example that is easy to implement:

(8.3) $\rho(u) = 1 / (1 + \exp(-\beta(k_L(u) - \theta)))$ (logistic link).

This converts enforcement strengths into an effective rigidity fraction without committing to any specific rule engine.

8.3 Meta-optimization objective (reuse existing scaffold)

We reuse the paper's main objective: maximize viability net of representation and enforcement costs.

(8.4) Maximize J(N, D, L; E) = V(L; E) − λ_N C(N) − λ_L Cost(L).

To make "controller" precise, treat L as containing adjustable enforcement parameters (including k_L(u)), and treat N and D as adaptable modules whose update rates may also be controlled.

A minimal control formulation:

(8.5) Choose k_L(·) to maximize E[$\sum_{t=0}^{T} \gamma^t R_t$] − $\lambda_N C(N) - \lambda_L \sum_{t=0}^{T} \gamma^t$ Cost_t(k_L(u_t)).

Subject to a stability / agreement constraint (the "band" claim):

(8.6) α_min ≤ ρ_eff,t · $\hat{\sigma}_{E,t}$ ≤ α_max.

This is the operational heart of the controller: it does not aim for maximal truth; it aims to maintain **viability within a band** by tuning enforcement strength so that anchoring is "just enough" relative to drift.

Optional but clean structural decomposition (ties back to Section 6).
Apply separate schedules to Name and Dao:

(8.7) k_L(u) = (k_name(u), k_policy(u), k_time(u)).
(8.8) $\vec{\rho}$(u) = (A_name(u), A_policy(u), A_time(u)) = $\vec{f}$(k_L(u)).

This is how you implement "A_name high, A_policy medium, A_time low" allocations while keeping ρ_eff near a mid-band.

8.4 Expected empirical signature

CLP makes a direct experimental prediction: for many drifting tasks, performance vs rigidity should exhibit a single interior maximum.

Let P(ρ) be any performance metric (long-run reward, survival rate, regret, coordination success). Then:

(8.9) P(ρ) is maximized at ρ* ∈ (0,1), often near 1/2 in the symmetric regime.

Equivalently, in terms of AB-fixness:

(8.10) P(A_B) peaks at an interior A_B* rather than at A_B=0 or A_B=1.

A "Half-Truth Controller" should therefore display two recognizable behaviors in simulation:

1. **Auto-centering:** when drift is moderate and budgets are adequate, the controller drives ρ_eff toward a mid-band.

2. **Regime response:** when $\hat{\sigma}_E$ increases (faster drift), the controller reduces k_L in drift-sensitive subsystems (more slack); when coordination stakes rise, it increases k_L in critical invariants (more anchoring).

Both behaviors are falsifiable: if a tuned controller is always improved by pushing rigidity to an extreme, then the CLP premise is wrong (or the task lacks the complementarity structure

assumed in Section 3).

---

9. Experiments (tight, publishable)

This section specifies a minimal but publishable experimental suite that directly tests the falsifiable claims made earlier:

- performance is **non-monotone** in rigidity,
- an interior optimum ρ* exists (often near 1/2 in symmetric regimes),
- AB-fixness provides a measurable rigidity proxy,
- anisotropic frame shifts induce "logic failures" that are resolved by Name updates (not by more rigid inference),
- and (optionally) a √n-style residual barrier appears in a synthetic task, mirroring the RH archetype.

9.1 Toy MDP suite: non-stationary gridworld

Goal. Demonstrate the basic CLP signature: best long-run performance occurs at mid-rigidity, not extremes.

Environment. A gridworld MDP with regime shifts. At each episode, the environment selects a regime $z_t \in \{1,\dots,K\}$ that changes one or more of:

- reward map $R_z(s,a)$,
- transition dynamics $P_z(s'|s,a)$,
- observation aliasing (partial observability), or
- goal location distribution.

A simple regime process:

(9.1) $z_{t+1} = z_t$ with prob $1-p_{switch}$, else sample $z_{t+1} \sim \mathrm{Unif}(\{1,\dots,K\}\{z_t\})$.

Define a drift scale proxy:

(9.2) $\sigma_E := p_{switch}$ (or a richer proxy, e.g., expected total-variation between P_z and $P_{z'}$).

Agents. Use a base learner (e.g., tabular Q-learning or a small neural policy) but vary only the rigidity control. Implement three regimes:

- Low rigidity: $\rho \approx 0$ (weak constraints, frequent re-naming / policy drift allowed).
- Mid rigidity: $\rho \approx 1/2$ (Half-Truth: stable core + adaptive sheath).
- High rigidity: $\rho \approx 1$ (strong constraints, slow re-naming, strict cross-time consistency).

If you operationalize rigidity via penalties:

(9.3) $\rho = f(k_L)$, monotone in k_L.

Metrics.

- long-run discounted return:

(9.4) Return = E[$\sum_{t=0}^{T} \gamma^t r_t$].

- stability vs adaptivity diagnostics:
    - naming drift (Section 6): $Drift_{name}(t,t+\Delta)$,
    - policy drift: $Drift_{policy}(t,t+\Delta)$,
    - constraint violation rate (if applicable).

Main prediction.
For moderate σ_E and finite budget, performance peaks at an interior rigidity:

(9.5) Return(ρ) achieves maximum at $\rho^* \in (0,1)$, typically near 1/2 in the symmetric setting.

You report:

- Return vs ρ curve (3–7 rigidity settings, not just 3 points),
- plus the three anchor points ($\rho \approx 0$, 1/2, 1) for interpretability.

9.2 Multi-agent alignment / disagreement tests

Goal. Validate AB-fixness as an empirical rigidity observable and show its tradeoff with adaptivity.

Setup.
Two agents A and B interact with either:

- the same environment stream, or
- two correlated streams with shared latent regimes but different local noise.

Implement variants:

- shared training data vs separate data,
- synchronized updates vs staggered updates (to probe time stability).

Measure AB-fixness directly:

(9.6) $\hat{A}_B(L) = E_{w \sim \pi_E}[I(N_A(w)=N_B(w) \wedge D_A(N_A(w))=D_B(N_B(w)))]$.

Estimate via samples $\{w_i\}$:

(9.7) $\hat{A}_B = (1/n) \sum_{i=1}^{n} I(N_A(w_i)=N_B(w_i) \wedge D_A(N_A(w_i))=D_B(N_B(w_i)))$.

Metrics.

- $\hat{A}_B$ as rigidity proxy,
- joint task success (team reward) vs $\hat{A}_B$,
- disagreement cost (if you charge arbitration cost).

Main predictions.

- Â_B increases monotonically with enforcement strength k_L.
- Task performance is maximized at intermediate Â_B when σ_E is moderate/high (too rigid → slow adaptation; too loose → poor coordination).

(9.8) Performance(Â_B) is unimodal with maximum at Â_B* ∈ (0,1).

9.3 Anisotropic-frame perturbation experiment

Goal. Demonstrate "absolute logic can be a coordinate illusion" and test the recovery advantage of adaptive Name-updating.

Environment / representation.
Let observations embed to $x \in R^3$. Initially use isotropic metric:

(9.9) $G_0 = \mathrm{diag}(1,1,1)$.

At time $t = t^*$, perturb the representational geometry to anisotropic:

(9.10) $G_1 = \mathrm{diag}(g_1,g_2,g_3)$, with (g_1,g_2,g_3) not all equal.

This can be implemented as:

- a literal change in the distance metric used by the naming module (nearest-prototype assignment), or
- a change in feature scaling upstream (equivalent).

Agents.
Compare two classes:

1. **Fixed-logic / fixed-name agents**: hold N fixed (or allow only minimal updates) and attempt to preserve constraints by increasing enforcement when errors spike.
2. **Adaptive Name-updating agents**: allow N (and/or Φ, the embedding) to update to restore a compact invariant core under the new frame; maintain a Half-Truth split so residual absorbs the shock.

Measures.

- pre/post perturbation performance drop:

(9.11) ΔReturn = Return_pre − Return_post.

- recovery time (episodes to regain ≥η fraction of pre-shift performance).
- naming instability spike:

(9.12) Spike = Drift_name(t*−Δ, t*+Δ).

Main predictions.

- Fixed-name agents show persistent degradation or brittle failure after t* (mis-specified invariants).
- Adaptive Name-updating agents recover faster and stabilize a new invariant core.

(9.13) Recovery_time(adaptive) < Recovery_time(fixed), under comparable budgets.

9.4 RH-parable diagnostic (optional but elegant)

Goal. Provide a synthetic analogue of the "critical line barrier": a task where the best achievable generalization error has a √n-scale residual, and forcing error below that systematically destroys robustness.

Task template.
Let the target be a cumulative signal with martingale noise:

(9.14) $y_n = \mathrm{Trend}(n) + \sum_{t=1}^{n} \xi_t$, where $E[\xi_t]=0$ and $\mathrm{Var}(\xi_t)=\sigma^2$.

Then the irreducible residual behaves like:

(9.15) $\mathrm{SD}\left(\sum_{t=1}^{n} \xi_t \right) = \sigma\sqrt{n}$.

Construct a prediction game: the agent predicts y_n from partial information, and is penalized for both error and overfitting. Introduce a controller that tries to enforce a uniform bound tighter than √n, e.g., by adding hard constraints or excessive smoothing.

Manipulation.

- baseline: allow residual at √n scale,
- over-enforced: force residual below √n (by overly rigid smoothing / constraint enforcement).

Outcome.

- In-sample error may drop,
- out-of-distribution generalization breaks (because the system is now fitting noise as structure).

A clean measurement is a generalization gap:

(9.16) $\mathrm{Gap}(n) = \mathrm{Err}_{\mathrm{test}}(n) - \mathrm{Err}_{\mathrm{train}}(n)$.

Prediction.
Over-enforcing below the √n envelope increases Gap(n) sharply, supporting the narrative that there is a structural boundary between compressible trend and irreducible residual.

Reporting standard (so it reads publishable)

For each experiment, report:

- a small set of fixed hyperparameters (so reviewers can reproduce),
- a sweep over rigidity settings (k_L or ρ),
- means + confidence intervals across seeds,
- and the predicted qualitative signature (unimodal performance curve, recovery after frame shift, etc.).

10. Discussion: what this reframes, and what it leaves open

10.1 Why this is not "relativism"

The proposal "logic is engineered rigidity" can be misread as "anything goes." That is **not** what the Name–Dao–Logic framework permits.

(i) Viability can be multi-criteria, not a single scalar excuse.
The base framework explicitly allows viability to be vector-valued (truth-seeking, safety, fairness, robustness, etc.), so disagreement is structured as trade-offs—not as unconstrained relativism.

(10.1) V_vec(L; E) = (V_survival, V_truth, V_fairness, V_safety, …).

A system can then define admissibility via a meta-level weighting or constraint set:

(10.2) L is admissible iff V_vec(L; E) ∈ Ω_admissible.

Here Ω_admissible can encode non-negotiables ("never falsify evidence," "never violate safety constraints," "respect human rights constraints"), which live **above** local convenience.

(ii) Meta-logic is allowed (and often required).
When multiple logics L_1, L_2 appear locally viable but conflict, the framework permits an explicit arbitration layer L_meta that constrains which trade-offs are acceptable. This is not optional glue; it is the formal mechanism that blocks slide-into-relativism.

(10.3) L_meta : (L_1, L_2, …) ↦ decision over admissibility / priority / coexistence.

(iii) "Non-absolute" means "conditional on Name/Dao/frame," not "uncriticizable."
Because N (ontology) and D (policy) are explicit, you can audit *why* a logic behaves as it does: which invariants it enforces, where it allows slack, what it costs, and what it optimizes. That is a stronger notion of critique than metaphysical purity arguments, because it ties directly to operational objectives and failure modes.

10.2 Relationship to RH research

This paper is **not** a contribution to analytic number theory. The RH lens is used as a canonical example of a "critical boundary" phenomenon: a place where 1/2 is structurally special.

(i) RH as a canonical 1/2-boundary instance.
RH asserts nontrivial zeros lie on the critical line Re(s)=1/2; it is one of the most studied open problems in mathematics. (American Institute of Mathematics)
In the lens we adopt, the "1/2-line" functions as an archetypal boundary separating what can be globally controlled (trend) from what remains as an irreducible residual envelope (oscillation).

(ii) Partial results legitimize "partial anchoring" as a scientific object.
Even without RH, there are theorems proving that a **positive proportion** of zeros lie on the critical line. Conrey (1989) proves at least two-fifths are on the line. (American Institute of Mathematics)
Bui–Conrey–Young prove more than 41% are on the line. (American Institute of Mathematics)

In this paper, these serve as a mathematical parable for engineered rationality:

- the "ideal boundary" (ρ≈1/2 everywhere) may be unproven or infeasible, yet
- a system can still secure **provable / enforceable lower bounds** on anchoring (ρ ≥ 0.4, or 0.41+), while leaving the remaining degrees of freedom as adaptive slack.

(iii) What is (and is not) being claimed.

- Not claimed: RH ⇒ CLP, or CLP ⇒ RH.
- Claimed: RH and its partial results provide a clean, famous template where "1/2" is a structurally meaningful boundary and where "partial alignment with the boundary" is already a precise, incremental research object.

10.3 Open problems (future papers)

This paper deliberately stops at "principle + architecture + experiments." The following extensions are natural, and can each be its own paper.

*Open Problem A (theory): derive conditions for ρ ≈ 1/2 from an explicit model class.**
We introduced a complementarity-style objective (benefit requires both anchoring and slack). A rigorous version would specify a family of environments and agent classes (N, D, L), and prove that:

(10.4) ρ* = argmax_{ρ∈[0,1]} E[V(L_ρ; E)] satisfies ρ* ≈ 1/2 under symmetry conditions.

"Symmetry conditions" here should be stated concretely (e.g., symmetric penalties for under-coordination vs over-rigidity, stationary distribution regularity, balanced task loss geometry).

*Open Problem B (control law): closed-form relation between σ_E, budget, and ρ.**
The base Name–Dao–Logic paper already contains a *simple* coordination-game derivation where equilibrium AB-fixness A* becomes an explicit function of σ_E (and cost parameters). A next step is to generalize that result from toy games to broader AGI settings:

(10.5) ρ* = f(σ_E, B; θ_task, θ_agent),

and to identify which quantities are measurable online (σ̂_E, disagreement, enforcement overhead) such that a Half-Truth Controller can track ρ* in real time.

Open Problem C (geometry): when do frame shifts force "logic pivots," and how should systems localize them?
Section 7 showed that changing the representational metric changes naming boundaries and can create "contradictions" that are coordinate artifacts. A formal theory would characterize:

(10.6) minimal update ΔN (or ΔG) needed so that constraint-violation rate returns below threshold,

and prove conditions under which updating Name locally dominates "hardening logic globally."

Open Problem D (RH-parable formalization): characterize √·-barrier tasks in learning theory language.
The optional Experiment 9.4 gestures at tasks with an irreducible √n residual envelope. A publishable follow-up would define a task family where:

(10.7) best achievable generalization residual scales as Ω(√n) (up to logs),

and prove that forcing tighter uniform error bounds induces overfitting / loss of robustness—making the "critical boundary" claim mathematically crisp in an ML setting (without importing RH itself).

---

11. Conclusion

Logic, in the Name–Dao–Logic view, is not a timeless judge of propositions; it is an engineered **survival protocol**. Its "truth" is the portion of structure that remains worth enforcing once you account for drift, coordination, and enforcement cost.

(11.1) "Useful truth" = invariants enforced by L under budget and volatility constraints.

The Riemann Hypothesis lens supplies a canonical mathematical archetype for why **1/2** keeps reappearing as a natural boundary: the critical line Re(s)=1/2 is where a system's best-possible global control cleanly meets an irreducible residual (a "trend + fluctuation" boundary rather than total eliminability of oscillation).

Equally important, partial results already prove that "critical alignment" can be **nontrivial without being complete**: at least two-fifths of zeros on the critical line (and later >41%) show how a strong anchoring fraction can be secured even without a full global proof. In the engineered-rationality translation, this legitimizes **partial anchoring** as a first-class scientific and architectural target:

(11.2) $\rho_{min}(E,B) \le \rho \le \rho_{max}(E,B)$, with $\rho_{target} \approx 1/2$ when feasible.

Finally, the anisotropic 3-axis demonstration makes "non-absolute logic" concrete: what looks like universal logical necessity often depends on a hidden shared coordinate/metric choice in the naming layer. Change the frame, and "obvious" invariants—and thus what counts as contradiction—can change without any relativist leap.

11. Reference

Chung Leung Danny Yeung, (2025). Name, Dao, and Logic: A Scientific Field Theory of Engineered Rationality and Its AGI Implementation 名、道與邏輯：一套可實作的理性場論. https://osf.io/5bfkh/files/osfstorage/6935c47cbb5827a1378f1ca6

Appendix A. Notation & Blogger-ready equation conventions

A.1 One-page notation list

Sets / spaces

- **W**: world-instance space (raw situations, observations, contexts).
  w ∈ **W**: a single world instance.
- **X**: named/semantic state space (post-compression).
  x ∈ **X**: a named state.
- **A**: action space.
  a ∈ **A**: an action.
- **E**: environment (dynamics + reward + drift/volatility + data distribution assumptions).

Core maps

- **N : W → X**: Name map (compression / ontology / categorization).
- **D : X → A**: Dao map (policy / action selection).

Logic object

- **L**: a logic system / constraint layer.
  Often written as **L := (Σ, ⊢, U)** where:
    - **Σ**: symbol/statement system,
    - ⊢: inference relation,
    - **U**: utility / selection pressure / rule-weighting structure.

Return / viability

- **R(x_t, a_t; E)**: reward / survival increment at time t in environment E.
- **γ ∈ (0,1]**: discount factor.
- **T**: horizon length (finite or effective).
- **S(N,D;E)**: survival/return functional (trajectory total).
- **V(L;E)**: viability of logic L in environment E (expected return, possibly net of costs).

Agreement / rigidity

- **π_E**: environment-induced distribution over world instances (sampling distribution).
- **I(·)**: indicator function (1 if condition holds, else 0).
- **A_B(L)**: AB-fixness (cross-observer and/or cross-time agreement statistic).
- **ρ ∈ [0,1]**: anchoring fraction ("truth we actually enforce"), often **ρ := A_B(L)**.
- $\vec{\rho}$: vector rigidity profile, e.g. $\vec{\rho}$**:= (A_name, A_policy, A_time, …)**.
- **σ_E**: environment volatility/drift scale (any operational proxy you define).

Costs / enforcement

- **C(N)**: complexity/maintenance cost of naming (model size, description length, update overhead).
- **Cost(L)**: enforcement cost of logic (checking, arbitration, penalties, brittleness costs).
- **λ_N, λ_L ≥ 0**: tradeoff weights for costs in the objective.
- **k_L**: enforcement strength (penalty weight / strictness / "temperature"), possibly context-dependent: **k_L = k_L(u)**.

Anisotropic 3-axis geometry (Section 7)

- **Φ(w) ∈ R^3**: embedding/feature map of world instance w.
- **G = diag(g_1,g_2,g_3)**: diagonal metric weights (g_i>0).
- **||x||²_G = x^T G x**: metric-weighted norm.
- **Dist_G(a,b) = ||a−b||_G**: induced distance.
- **μ_k**: prototype/centroid for named category k (if using prototype naming).

A.2 Blogger-ready equation conventions

Rule A (single-line equations only).
Write every equation on one line. No LaTeX, no MathJax.

Rule B (tag every displayed equation).
Put the tag at the end, in parentheses: **(A.n)** or **(n.m)**.

Rule C (Unicode operators are preferred).
Use →, ∈, ≤, ≥, √, ∑, ⊕, ×, Δ, diag(·), argmin, argmax, E[·].

Rule D (AMS-style blocks, but still single-line equations).
You may format "Proposition / Proof / □" as plain text blocks; equations inside remain single-line.

A.3 Three formatting examples

Example 1: a definition

- N : W → X. (A.1)

Example 2: an expectation with a distribution and an indicator

- A_B(L) = E_{w ~ π_E}[I(N_A(w)=N_B(w) ∧ D_A(N_A(w))=D_B(N_B(w)))]. (A.2)

Example 3: a control / design rule

- ρ_design = clamp(1/2, ρ_min(E,B), ρ_max(E,B)). (A.3)

A.4 AMS-style block example (template)

Proposition A.1 (Symmetric complementarity optimum).
Assume J(ρ) = κ·ρ·(1−ρ) with κ>0. Then J is maximized at ρ* = 1/2.

Proof.

- dJ/dρ = κ·(1−2ρ). (A.4)
- Setting dJ/dρ = 0 gives ρ* = 1/2. (A.5)
- d²J/dρ² = −2κ < 0, so the point is a maximum. (A.6)
  □

Appendix B. RH facts used (minimal, citation-friendly)

B.1 RH statement (critical line)

Let ζ(s) be the Riemann zeta function, and call a zero "nontrivial" if it lies in the critical strip 0 < Re(s) < 1.

(B.1) RH: ζ(s)=0 and 0<Re(s)<1 ⇒ Re(s)=1/2. (American Institute of Mathematics)

This is the only RH fact the paper directly needs: **the critical line Re(s)=1/2 is the conjectured universal location of all nontrivial zeros**. (American Institute of

Mathematics)

B.2 √x-style consequence (prime-counting error envelope)

Let $\pi(x)$ be the prime-counting function and $\mathrm{li}(x)$ the logarithmic integral approximation.

A standard consequence of RH is a "square-root–scale" global error bound (up to log factors):

(B.2) $\pi(x) - \mathrm{li}(x) = O(\sqrt{x} \cdot \log x)$. (American Mathematical Society)

Interpretation used in this paper: the √x envelope is a canonical signature of **"structured trend + irreducible residual"**—the trend is compressible (li(x)), while the residual cannot be uniformly eliminated beyond the √· scale without stronger structural claims about zeros.

(For a more general statement tying the error exponent to the supremum of real parts of zeros, see the "Consequences" discussion that summarizes $\pi(x) - \mathrm{li}(x) = O(x^{\beta} \log x)$ where β reflects how far right zeros can be. (Wikipedia))

B.3 "≥2/5" and ">41%" on the critical line (partial criticality)

Define counting functions up to height T:

(B.3) $N(T) := \#\{ \rho = \beta+i\gamma : \zeta(\rho)=0, 0<\gamma\le T, 0<\beta<1 \}$.
(B.4) $N_0(T) := \#\{ \rho = 1/2+i\gamma : \zeta(\rho)=0, 0<\gamma\le T \}$.

Define the lower-density of zeros on the critical line:

(B.5) $\kappa := \liminf_{T\to\infty} N_0(T) / N(T)$.

Conrey (1989) proved that at least two-fifths of the zeros are (in fact, simple and) on the critical line, which yields:

(B.6) $\kappa \ge 2/5$. (American Institute of Mathematics)

Bui–Conrey–Young (2011) proved a stronger bound, commonly summarized as "more than 41%" on the critical line:

(B.7) $\kappa > 0.41$. (American Institute of Mathematics)

B.4 How these facts are used in the paper (one sentence)

- RH supplies the **canonical 1/2 boundary** template (all nontrivial zeros on Re(s)=1/2). (American Institute of Mathematics)
- The √x-style consequence supplies the **"critical boundary = best-possible global control up to irreducible residual"** archetype. (American Mathematical Society)
- The ≥2/5 and >41% results justify **partial anchoring** as a serious, provable object even without full RH—mirroring the paper's "non-absolute but engineerable truth" theme. (American Institute of Mathematics)

Appendix C. Experimental protocols

This appendix specifies **measurement recipes** (not model choices) so results are reproducible

even if the underlying learner changes.

C.1 Common experimental scaffolding (applies to all experiments)

Seeds and runs

- Use **S** random seeds per condition (recommend S ≥ 20).
- Each seed produces **Eps** evaluation episodes (recommend Eps ≥ 200 for toy MDPs).

Episode data log (minimum)
For each timestep t in each episode, log:

- world instance: w_t (or a stable ID / hash if large),
- embedding: Φ(w_t) (if used),
- named state: x_t = N(w_t),
- action: a_t = D(x_t),
- reward: r_t,
- regime label: z_t (if the environment has regimes),
- enforcement strength: k_L(u_t) (or rigidity proxy ρ_t).

C.2 AB-fixness measurement (A_B)

C.2.1 "Hard equality" AB-fixness (baseline)

Given paired agents A and B evaluated on the *same sampled world instances* {w_i} drawn from π_E:

(C.1) I_AB(w) = I(N_A(w)=N_B(w) ∧ D_A(N_A(w))=D_B(N_B(w))).
(C.2) Â_B = (1/n) ∑_{i=1}^{n} I_AB(w_i).

Where n is the number of sampled instances used for estimation.

Protocol

1. Collect n world instances {w_i} from evaluation rollouts (recommended n ≥ 10,000 across episodes).
2. For each w_i, query both agents for:
    - x_A,i = N_A(w_i), x_B,i = N_B(w_i),
    - a_A,i = D_A(x_A,i), a_B,i = D_B(x_B,i).
3. Compute I_AB(w_i) and average to get Â_B.

C.2.2 "Softened" AB-fixness (recommended for continuous states)

If exact equality is too strict, define similarity thresholds.

Let sim_X(x,x′) ∈ [0,1] be a naming similarity (e.g., cosine similarity of embeddings, or equality of cluster IDs), and sim_A(a,a′) ∈ [0,1] be an action similarity (e.g., same discrete action, or small L2 distance for continuous actions).

(C.3) I_AB^soft(w) = I(sim_X(N_A(w),N_B(w)) ≥ τ_X ∧ sim_A(D_A(N_A(w)),D_B(N_B(w))) ≥ τ_A).
(C.4) Â_B^soft = (1/n) ∑_{i=1}^{n} I_AB^soft(w_i).

Report both Â_B (hard) if defined and Â_B^soft (soft) with thresholds (τ_X, τ_A).

C.2.3 Decomposed agreement (Name-only vs Policy-only)

To diagnose where disagreement comes from:

(C.5) Â_name = (1/n) ∑ I(N_A(w_i)=N_B(w_i)).
(C.6) Â_policy = (1/n) ∑ I(D_A(N_A(w_i))=D_B(N_B(w_i))).

And optional time-stability for one agent:

(C.7) Â_time(Δ) = (1/n) ∑ I(N_t(w_i)=N_{t+Δ}(w_i)).

C.3 Environment volatility / drift proxy (σ_E)

σ_E should be **defined per environment** but measured consistently.

C.3.1 Regime-switch proxy (when regimes exist)

If the environment has explicit regime z_t and a switching probability:

(C.8) σ_E := p_switch.

Empirically estimate from logs:

(C.9) σ̂_E = (1/(T_total−1)) ∑*{t=1}^{T_total−1} I(z*{t+1} ≠ z_t).

C.3.2 Transition-drift proxy (model-free, when regimes are hidden)

Estimate drift via distribution shift in observed transitions.

Let f_t be a feature summary of recent experience over a window W (e.g., histogram of (x,a,x′), or embedding statistics).

(C.10) f_t = Summary({ (x_i,a_i,x′*i) }*{i=t−W+1}^{t}).
(C.11) σ̂_E(t;Δ) = D_div(f_t, f_{t+Δ}).

Where D_div can be Jensen–Shannon divergence (recommended for boundedness) or total variation if histograms.

C.3.3 Reward-drift proxy (if dynamics stable but goals change)

(C.12) g_t = Summary({ r_i }*{i=t−W+1}^{t}).
(C.13) σ̂_E(t;Δ) = | E[g_t] − E[g{t+Δ}] | (or a divergence on reward histograms).

Reporting rule: Always state which σ̂_E definition is used, window size W, and Δ.

C.4 Rigidity proxy (ρ) and mapping from enforcement strength

If you directly control ρ, log it. If you control penalties k_L, define a monotone link so plots are comparable.

(C.14) ρ_t = f(k_L(u_t)), with f′ ≥ 0 and ρ_t ∈ [0,1].

A concrete default link (if you need one):

(C.15) ρ_t = 1 / (1 + exp(−β(k_L(u_t) − θ))).

Empirical alternative (preferred): treat AB-fixness itself as ρ.

(C.16) ρ := Â_B (or Â_B^soft).

C.5 Outcome metrics (what you optimize / compare)

C.5.1 Return and success

Discounted return per episode e:

(C.17) G_e = ∑{t=0}^{T_e} γ^t r{e,t}.
(C.18) Return = (1/Eps) ∑_{e=1}^{Eps} G_e.

If tasks have success/failure:

(C.19) SuccessRate = (1/Eps) ∑_{e=1}^{Eps} I(success_e).

C.5.2 Regret (optional but clean)

If an optimal benchmark is available (or best-in-sweep baseline):

(C.20) Regret = (1/Eps) ∑_{e=1}^{Eps} (G*_e − G_e).

C.5.3 Coordination cost / disagreement cost (multi-agent)

If arbitration cost is modeled:

(C.21) Cost_coord = (1/n) ∑_{i=1}^{n} I(N_A(w_i)≠N_B(w_i) ∨ a_A,i≠a_B,i).

Or weighted by a unit cost c_coord:

(C.22) Cost_coord = (c_coord/n) ∑ I_AB_conflict(w_i).

C.5.4 Recovery after perturbation (anisotropic shift experiment)

Let t* be the intervention time. Define pre-shift baseline as average performance over a window [t*−H, t*).

(C.23) Perf_pre = (1/H) ∑{t=t*−H}^{t*−1} r_t.
(C.24) Perf_post(t) = (1/H) ∑{i=t}^{t+H−1} r_i.

Define recovery time to reach fraction η of baseline:

(C.25) T_recover(η) = min{ t ≥ t* : Perf_post(t) ≥ η·Perf_pre }.

Report also immediate drop:

(C.26) Drop = Perf_pre – Perf_post(t*).

C.6 Statistical reporting (minimal but sufficient)

For any metric M (Return, Â_B, σ̂_E, etc.), computed per seed s:

(C.27) M̄ = (1/S) ∑_{s=1}^{S} M_s.
(C.28) SE(M) = SD(M_s) / √S.

Report either:

- mean ± 1.96·SE (approx 95% CI), or
- bootstrap CI across seeds (state method).

Unimodality test (for CLP signature)
When sweeping rigidity settings {ρ_j}, report the argmax and show non-monotonicity:

(C.29) ρ̂* = argmax_{j} Return(ρ_j).

Optionally fit a quadratic locally around the peak to confirm an interior maximum (state fit domain).

C.7 Minimum plots/tables to include in the paper (no extra fluff)

1. Return vs ρ (or vs Â_B) with CI.
2. Â_B vs k_L (to show monotonic controllability).
3. σ̂_E over time (for non-stationary tasks).
4. Perturbation experiment: performance trace with t* marked + recovery time distribution.

That’s enough to make the experimental section tight and reviewer-friendly.

© 2025 Danny Yeung. All rights reserved. 版权所有 不得转载

Disclaimer

This book is the product of a collaboration between the author and OpenAI's GPT-5.2, Google's Gemini 3 Pro, X's Grok 4.1 language model. While every effort has been made to ensure accuracy, clarity, and insight, the content is generated with the assistance of artificial intelligence and may contain factual, interpretive, or mathematical errors. Readers are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.